
5.4.0 End of Chapter Problems

Problem 1

Consider two random variables X and Y with joint PMF given in Table 5.4

Joint PMF of X and Y in Problem 1

	$Y = 1$	$Y = 2$
$X = 1$	$\frac{1}{3}$	$\frac{1}{12}$
$X = 2$	$\frac{1}{6}$	0
$X = 4$	$\frac{1}{12}$	$\frac{1}{3}$

- Find $P(X \leq 2, Y > 1)$.
- Find the marginal PMFs of X and Y .
- Find $P(Y = 2|X = 1)$.
- Are X and Y independent?

Problem 2

Let X and Y be as defined in Problem 1. I define a new random variable $Z = X - 2Y$.

- Find the PMF of Z .
- Find $P(X = 2|Z = 0)$.

Problem 3

A box contains two coins: a regular coin and a biased coin with $P(H) = \frac{2}{3}$. I choose a coin at random and toss it once. I define the random variable X as a Bernoulli random variable associated with this coin toss, i.e., $X = 1$ if the result of the coin toss is heads and $X = 0$ otherwise. Then I take the remaining coin in the box and toss it once. I define the random variable Y as a Bernoulli random variable associated with the second coin toss. Find the joint PMF of X and Y . Are X and Y independent?

Problem 4

Consider two random variables X and Y with joint PMF given by

$$P_{XY}(k, l) = \frac{1}{2^{k+l}}, \text{ for } k, l = 1, 2, 3, \dots$$

- Show that X and Y are independent and find the marginal PMFs of X and Y .
- Find $P(X^2 + Y^2 \leq 10)$.

Problem 5

Let X and Y be as defined in Problem 1. Also, suppose that we are given that $Y = 1$.

- Find the conditional PMF of X given $Y = 1$. That is, find $P_{X|Y}(x|1)$.
 - Find $E[X|Y = 1]$.
 - Find $Var(X|Y = 1)$.
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Problem 6

The number of customers visiting a store in one hour has a Poisson distribution with mean $\lambda = 10$. Each customer is a female with probability $p = \frac{3}{4}$ independent of other customers. Let X be the total number of customers in a one-hour interval and Y be the total number of female customers in the same interval. Find the joint PMF of X and Y .

Problem 7

Let $X \sim \text{Geometric}(p)$. Find $\text{Var}(X)$ as follows: Find EX and EX^2 by conditioning on the result of the first "coin toss", and use $\text{Var}(X) = EX^2 - (EX)^2$.

Problem 8

Let X and Y be two independent $\text{Geometric}(p)$ random variables. Find $E\left[\frac{X^2 + Y^2}{XY}\right]$.

Problem 9

Consider the set of points in the set C :

$$C = \{(x, y) | x, y \in \mathbb{Z}, x^2 + |y| \leq 2\}.$$

Suppose that we pick a point (X, Y) from this set completely at random. Thus, each point has a probability of $\frac{1}{11}$ of being chosen.

- Find the joint and marginal PMFs of X and Y .
 - Find the conditional PMF of X given $Y = 1$.
 - Are X and Y independent?
 - Find $E[XY^2]$.
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Problem 10

Consider the set of points in the set C :

$$C = \{(x, y) | x, y \in \mathbb{Z}, x^2 + |y| \leq 2\}.$$

Suppose that we pick a point (X, Y) from this set completely at random. Thus, each point has a probability of $\frac{1}{11}$ of being chosen.

- Find $E[X|Y = 1]$.
 - Find $\text{Var}(X|Y = 1)$.
 - Find $E[X||Y| \leq 1]$.
 - Find $E[X^2||Y| \leq 1]$.
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Problem 11

The number of cars being repaired at a small repair shop has the following PMF:

$$P_N(n) = \begin{cases} \frac{1}{8} & \text{for } n = 0 \\ \frac{1}{8} & \text{for } n = 1 \\ \frac{1}{4} & \text{for } n = 2 \\ \frac{1}{2} & \text{for } n = 3 \\ 0 & \text{otherwise} \end{cases}$$

Each car that is being repaired is a four-door car with probability $\frac{3}{4}$ and a two-door car with probability $\frac{1}{4}$, independently from other cars and independently from the number of cars being repaired. Let X be the number of four-door cars and Y be the number of two-door cars currently being repaired.

- Find the marginal PMFs of X and Y .
 - Find the joint PMF of X and Y .
 - Are X and Y independent?
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Problem 12

Let X and Y be two independent random variables with PMFs

$$P_X(k) = P_Y(k) = \begin{cases} \frac{1}{5} & \text{for } x = 1, 2, 3, 4, 5 \\ 0 & \text{otherwise} \end{cases}$$

Define $Z = X - Y$. Find the PMF of Z .

Problem 13

Consider two random variables X and Y with joint PMF given in Table 5.5

Table 5.5: Joint PMF of X and Y in Problem 13

	$Y = 0$	$Y = 1$	$Y = 2$
$X = 0$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{8}$
$X = 1$	$\frac{1}{8}$	$\frac{1}{6}$	$\frac{1}{4}$

Define the random variable Z as $Z = E[X|Y]$.

- Find the Marginal PMFs of X and Y .
- Find the conditional PMF of X , given $Y = 0$ and $Y = 1$, i.e., find $P_{X|Y}(x|0)$ and $P_{X|Y}(x|1)$.
- Find the PMF of Z .
- Find EZ , and check that $EZ = EX$.
- Find $\text{Var}(Z)$.

Problem 14

Let X , Y , and $Z = E[X|Y]$ be as in Problem 13. Define the random variable V as $V = \text{Var}(X|Y)$.

- Find the PMF of V .
- Find EV .
- Check that $\text{Var}(X) = EV + \text{Var}(Z)$.

Problem 15

Let N be the number of phone calls made by the customers of a phone company in a given hour. Suppose that $N \sim \text{Poisson}(\beta)$, where $\beta > 0$ is known. Let X_i be the length of the i 'th phone call, for $i = 1, 2, \dots, N$. We assume X_i 's are independent of each other and also independent of N . We further assume

$$X_i \sim \text{Exponential}(\lambda),$$

where $\lambda > 0$ is known. Let Y be the sum of the lengths of the phone calls, i.e.,

$$Y = \sum_{i=1}^N X_i.$$

Find EY and $\text{Var}(Y)$.

Problem 16

Let X and Y be two jointly continuous random variables with joint PDF

$$f_{XY}(x, y) = \begin{cases} \frac{1}{2}e^{-x} + \frac{cy}{(1+x)^2} & 0 \leq x, 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- Find the constant c .
- Find $P(0 \leq X \leq 1, 0 \leq Y \leq \frac{1}{2})$.
- Find $P(0 \leq X \leq 1)$.

Problem 17

Let X and Y be two jointly continuous random variables with joint PDF

$$f_{XY}(x, y) = \begin{cases} e^{-xy} & 1 \leq x \leq e, y > 0 \\ 0 & \text{otherwise} \end{cases}$$

- Find the marginal PDFs, $f_X(x)$ and $f_Y(y)$.
- Write an integral to compute $P(0 \leq Y \leq 1, 1 \leq X \leq \sqrt{e})$.

Problem 18

Let X and Y be two jointly continuous random variables with joint PDF

$$f_{XY}(x, y) = \begin{cases} \frac{1}{4}x^2 + \frac{1}{6}y & -1 \leq x \leq 1, 0 \leq y \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

- Find the marginal PDFs, $f_X(x)$ and $f_Y(y)$.
- Find $P(X > 0, Y < 1)$.
- Find $P(X > 0 \text{ or } Y < 1)$.
- Find $P(X > 0 | Y < 1)$.
- Find $P(X + Y > 0)$.

Problem 19

Let X and Y be two jointly continuous random variables with joint CDF

$$F_{XY}(x, y) = \begin{cases} 1 - e^{-x} - e^{-2y} + e^{-(x+2y)} & x, y > 0 \\ 0 & \text{otherwise} \end{cases}$$

- Find the joint PDF, $f_{XY}(x, y)$.
- Find $P(X < 2Y)$.
- Are X and Y independent?

Problem 20

Let $X \sim N(0, 1)$.

- Find the conditional PDF and CDF of X given $X > 0$.
- Find $E[X | X > 0]$.
- Find $\text{Var}(X | X > 0)$.

Problem 21

Let X and Y be two jointly continuous random variables with joint PDF

$$f_{XY}(x, y) = \begin{cases} x^2 + \frac{1}{3}y & -1 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

For $0 \leq y \leq 1$, find the following:

- The conditional PDF of X given $Y = y$.
- $P(X > 0 | Y = y)$. Does this value depend on y ?
- Are X and Y independent?

Problem 22

Let X and Y be two jointly continuous random variables with joint PDF

$$f_{XY}(x, y) = \begin{cases} \frac{1}{2}x^2 + \frac{2}{3}y & -1 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Find $E[Y|X = 0]$ and $\text{Var}(Y|X = 0)$.

Problem 23

Consider the set

$$E = \{(x, y) | |x| + |y| \leq 1\}.$$

Suppose that we choose a point (X, Y) uniformly at random in E . That is, the joint PDF of X and Y is given by

$$f_{XY}(x, y) = \begin{cases} c & (x, y) \in E \\ 0 & \text{otherwise} \end{cases}$$

- Find the constant c .
 - Find the marginal PDFs $f_X(x)$ and $f_Y(y)$.
 - Find the conditional PDF of X given $Y = y$, where $-1 \leq y \leq 1$.
 - Are X and Y independent?
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Problem 24

Let X and Y be two independent $Uniform(0, 2)$ random variables. Find $P(XY < 1)$.

Problem 25

Suppose $X \sim Exponential(1)$ and given $X = x$, Y is a uniform random variable in $[0, x]$, i.e.,

$$Y|X = x \sim Uniform(0, x),$$

or equivalently

$$Y|X \sim Uniform(0, X).$$

- Find EY .
 - Find $\text{Var}(Y)$.
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Problem 26

Let X and Y be two independent $Uniform(0, 1)$ random variables. Find

- $E[XY]$
 - $E[e^{X+Y}]$
 - $E[X^2 + Y^2 + XY]$
 - $E[Ye^{XY}]$
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Problem 27

Let X and Y be two independent $Uniform(0, 1)$ random variables, and $Z = \frac{X}{Y}$. Find the CDF and PDF of Z .

Problem 28

Let X and Y be two independent $N(0, 1)$ random variables, and $U = X + Y$.

- Find the conditional PDF of U given $X = x$, $f_{U|X}(u|x)$.
- Find the PDF of U , $f_U(u)$.
- Find the conditional PDF of X given $U = u$, $f_{X|U}(x|u)$.

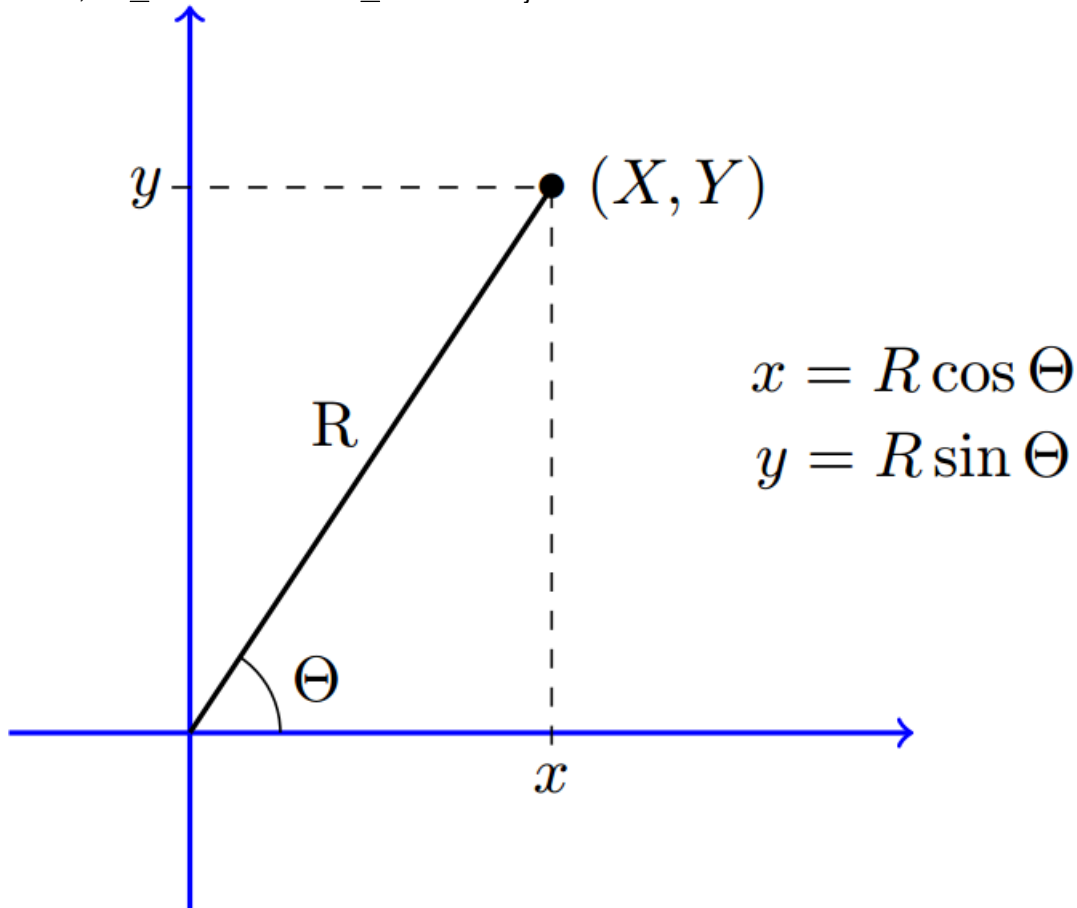
d. Find $E[X|U = u]$, and $Var(X|U = u)$.

Problem 29

Let X and Y be two independent standard normal random variables. Consider the point (X, Y) in the $x - y$ plane. Let (R, Θ) be the corresponding polar coordinates as shown in Figure 5.11. The inverse transformation is given by

$$\begin{cases} X = R \cos \Theta \\ Y = R \sin \Theta \end{cases}$$

where, $R \geq 0$ and $-\pi < \Theta \leq \pi$. Find the joint PDF of R and Θ . Show that R and Θ are independent.



Problem 30

In Problem 29, suppose that X and Y are independent $Uniform(0, 1)$ random variables. Find the joint PDF of R and Θ . Are R and Θ independent?

Problem 31

Consider two random variables X and Y with joint PMF given in Table 5.6.

Table 5.6: Joint PMF of X and Y in Problem 31

	$Y = 0$	$Y = 1$	$Y = 2$
$X = 0$	$\frac{1}{6}$	$\frac{1}{4}$	$\frac{1}{8}$
$X = 1$	$\frac{1}{8}$	$\frac{1}{6}$	$\frac{1}{6}$

Find $Cov(X, Y)$ and $\rho(X, Y)$.

Problem 32

Let X and Y be two independent $N(0, 1)$ random variable and

$$\begin{aligned} Z &= 11 - X + X^2Y, \\ W &= 3 - Y. \end{aligned}$$

Find $\text{Cov}(Z, W)$.

Problem 33

Let X and Y be two random variables. Suppose that $\sigma_X^2 = 4$, and $\sigma_Y^2 = 9$. If we know that the two random variables $Z = 2X - Y$ and $W = X + Y$ are independent, find $\text{Cov}(X, Y)$ and $\rho(X, Y)$.

Problem 34

Let $X \sim \text{Uniform}(1, 3)$ and $Y|X \sim \text{Exponential}(X)$. Find $\text{Cov}(X, Y)$.

Problem 35

Let X and Y be two independent $N(0, 1)$ random variable and

$$\begin{aligned} Z &= 7 + X + Y, \\ W &= 1 + Y. \end{aligned}$$

Find $\rho(Z, W)$.

Problem 36

Let X and Y be jointly normal random variables with parameters $\mu_X = -1$, $\sigma_X^2 = 4$, $\mu_Y = 1$, $\sigma_Y^2 = 1$, and $\rho = -\frac{1}{2}$.

- Find $P(X + 2Y \leq 3)$.
 - Find $\text{Cov}(X - Y, X + 2Y)$.
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Problem 37

Let X and Y be jointly normal random variables with parameters $\mu_X = 1$, $\sigma_X^2 = 4$, $\mu_Y = 1$, $\sigma_Y^2 = 1$, and $\rho = 0$.

- Find $P(X + 2Y > 4)$.
 - Find $E[X^2Y^2]$.
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Problem 38

Let X and Y be jointly normal random variables with parameters $\mu_X = 2$, $\sigma_X^2 = 4$, $\mu_Y = 1$, $\sigma_Y^2 = 9$, and $\rho = -\frac{1}{2}$.

- Find $E[Y|X = 3]$.
 - Find $\text{Var}(Y|X = 2)$.
 - Find $P(X + 2Y \leq 5|X + Y = 3)$.
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